

IT Operations Guide

TGIR QuantLib Tools

TG Investments and Research

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1 Purpose

This guide is for IT and operations staff responsible for runtime support, access control, incident response, backups, and production hygiene.

2 Runtime Summary

The application is a single Flask service with no database. Operational state is limited to:

- the deployed code,
- the Python virtual environment,
- the local `.env` file,
- logs and service definitions,
- the generated debug CSV under the app's debug directory, and
- standard operating-system service state.

3 Minimum Environment Requirements

- Linux host or local macOS workstation.
- Python 3 with the ability to create `.venv/`.
- QuantLib Python package installation support.
- Apache reverse proxy and a local WSGI service for production.
- Read-write access to the debug output path.

4 Secrets And Configuration

4.1 Required Variables

Variable	Purpose
FLASK_SECRET_KEY	Signs Flask sessions.
APP_LOGIN_USERNAME	Single application login username.
APP_LOGIN_PASSWORD or APP_LOGIN_PASSWORD_HASH	Login credential.
PORT or FLASK_RUN_PORT	Local port override if required.

4.2 Storage Rules

Do not commit secrets. Store production secrets in a server-side `.env` file outside Git-controlled directories, and use least-privilege filesystem permissions.

5 Operational Tasks

5.1 Start And Stop

Local development uses `./venv/bin/python app.py`. Production should use a systemd-managed WSGI service rather than the Flask development server.

5.2 Health Check

Use `/health` as the first smoke check after restart or deployment. It should return a JSON document with `"ok": true`.

5.3 Logs

Primary production logs should come from:

- systemd journal for the WSGI process,
- Apache access and error logs, and
- CI/CD workflow logs for deployment history.

6 Security Checklist

- Keep `FLASK_SECRET_KEY` strong and unique.
- Run behind HTTPS in production.
- Restrict shell access to the droplet.
- Patch the host OS and Apache packages regularly.
- Use a dedicated service account when practical.
- Keep the `.env` file outside world-readable paths.

7 Backup And Recovery

There is no database to back up. Recovery focuses on:

- code releases,
- the production `.env` file,
- Apache and systemd unit templates, and
- any shared debug or log directories kept outside the current release.

The deployment guide uses a `current` symlink plus timestamped release directories so rollback can be a symlink change followed by a service restart.

8 Troubleshooting

Symptom	Operational response
Service will not start	Confirm Python package install, <code>.env</code> presence, and systemd working directory.
<code>/health</code> fails	Check local bind port, WSGI service, and Apache proxy settings.
Login fails in production	Verify the deployed <code>.env</code> credential values and file permissions.
Disk growth	Clear old release directories, old logs, and generated artifacts according to retention policy.

9 Reference Files

Operations should know these files:

- `wsgi.py`,
- `requirements-production.txt`,
- `deploy/systemd/tgir-quantlib-tools.service.template`,
- `deploy/apache/tgir-quantlib-tools.conf.template`, and
- `.github/workflows/deploy_digitalocean.yml`.